Red Hat Summit

Demystifying Quantization

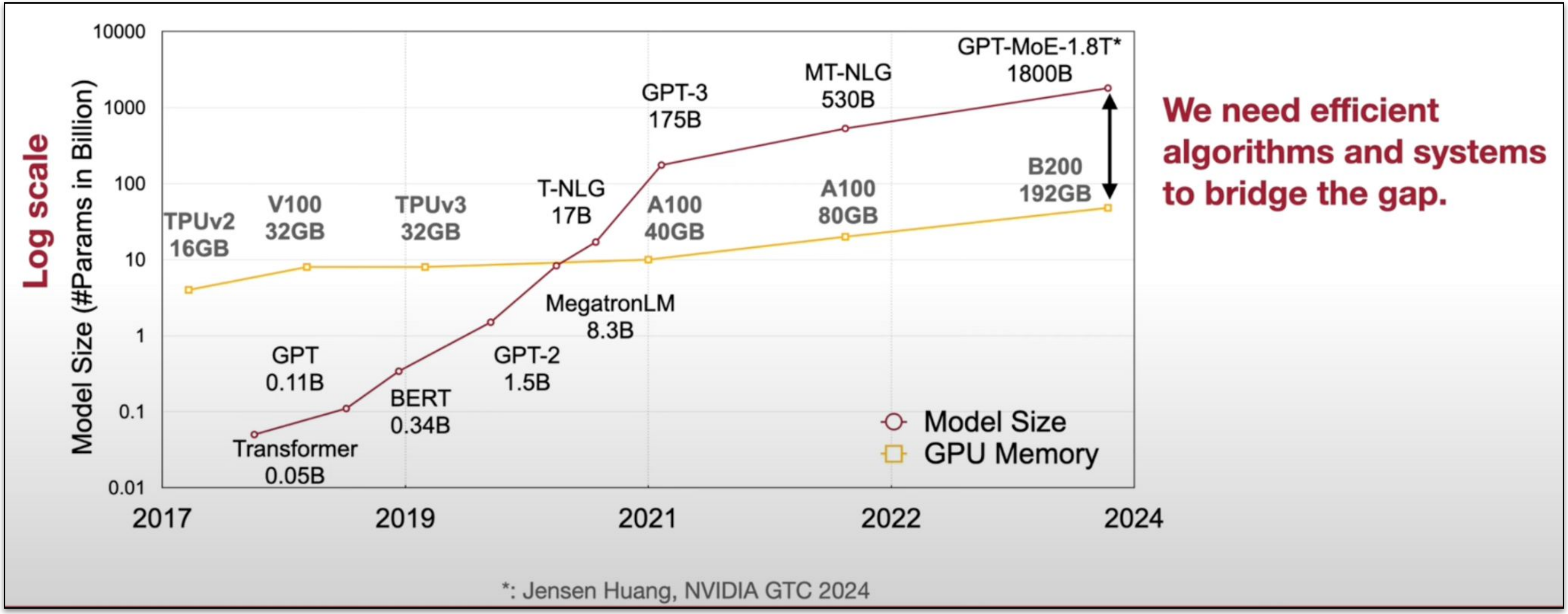
Accelerating LLM inference with Red Hat AI

Brian Dellabetta
Senior Machine Learning Engineer
Red Hat AI

Kyle Sayers
Senior Machine Learning Engineer
Red Hat AI

Red Hat

The AI Scaling Challenge



Red Hat

Costs and Risks Grow with Larger Models

💰 GPUs and Infrastructure

- GPU hardware costs
- Multi-GPU/ node deployment

⚡ Energy and Carbon Footprint

- Increased power consumption
- Increased data center land and water usage

📉 Tradeoffs in User Experience

- Response latency
- Request throughput
- Context length (KV cache)

❗ Risk of Model Obsolescence

- Larger models require more investment

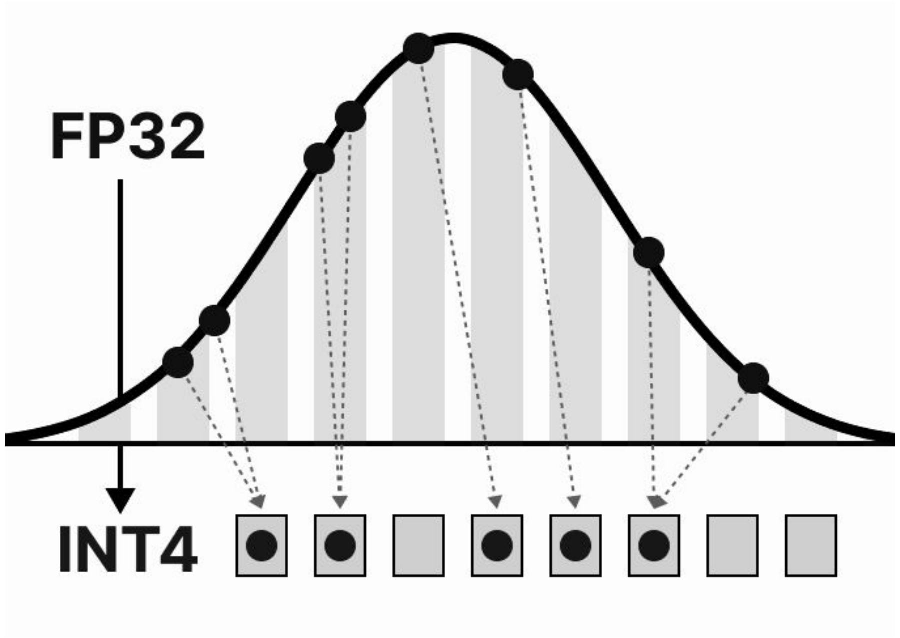
Red Hat

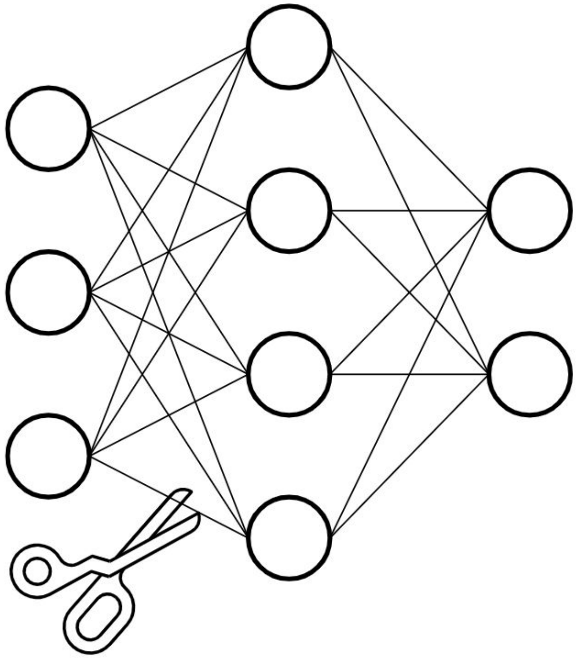
How can AI providers address the risks and costs of AI at scale?

Red Hat

Large Language Model Optimization

Quantization & Sparsification





Quantization, Sparsification, and Compression

- **Quantization:** Reduces model memory by representing weights and activations (i.e model layer inputs/outputs) using a lower bit representation (e.g int8). Allows models to use less storage. To quantize values to their new, low precision range:

$$\mathbf{X}_{\text{quant}} = \text{round}\left(\text{scale} \cdot \mathbf{X} + \text{zeropoint}\right)$$

$$\mathbf{X}_{\text{dequant}} = \frac{\mathbf{X}_{\text{quant}} - \text{zeropoint}}{\text{scale}}$$

- **Sparsification:** Certain model weight values are zeroed out. This can be done in a particular pattern, such as 2 of 4 where 2 out of every 4 values within a model weight tensor are set to 0.

- **Compression:** Saving the model in a reduced size (with minimal impact to model accuracy)

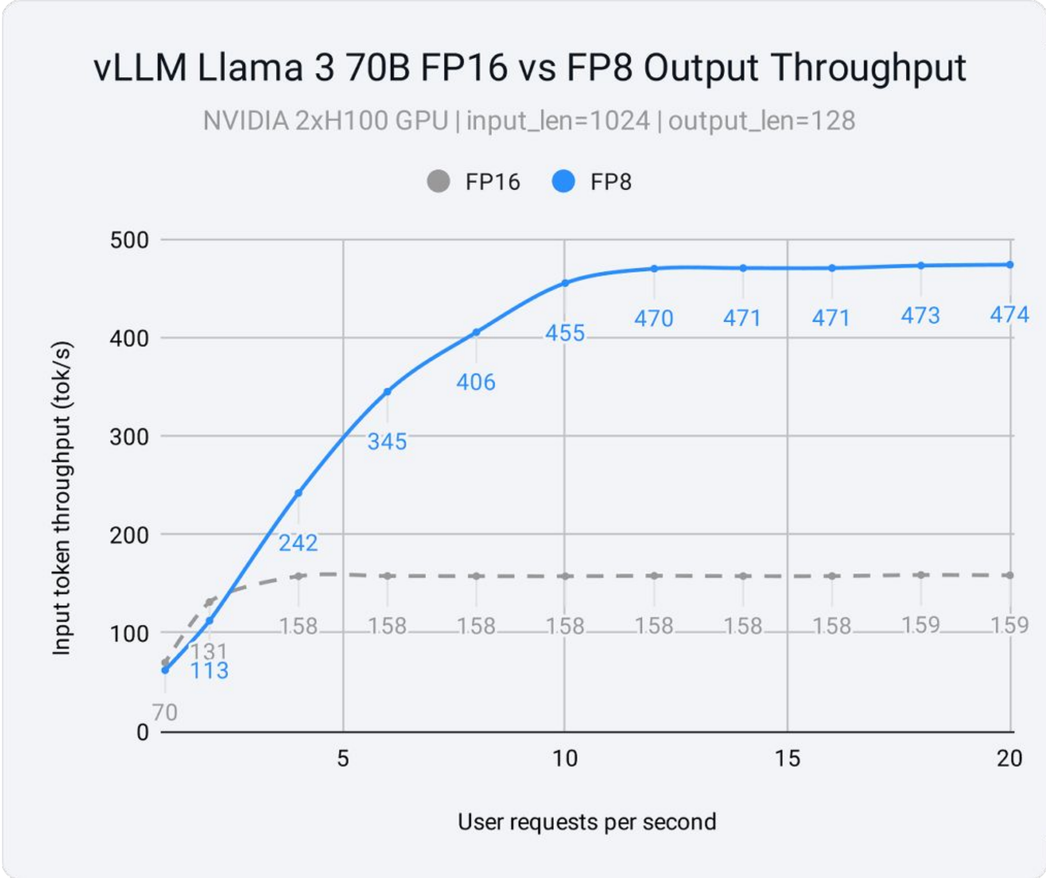
Why Optimize LLMs?

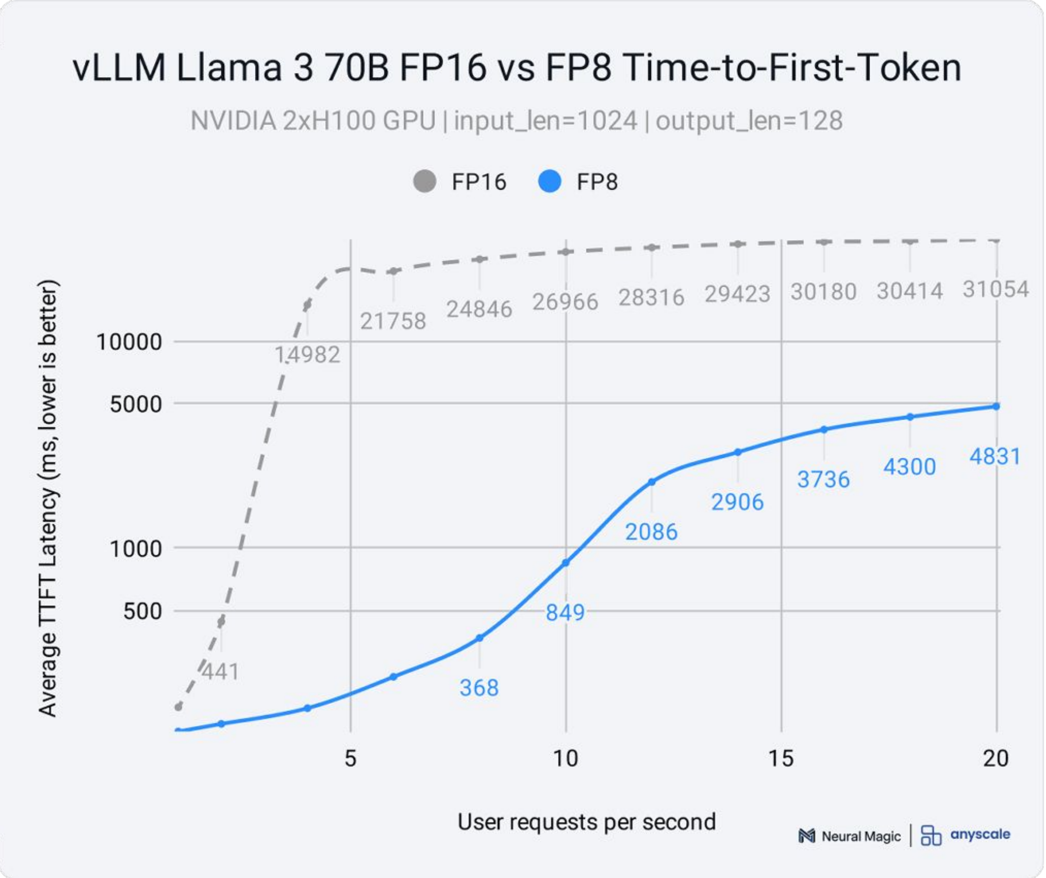
Reduced hardware requirements!

Optimization	Params size (GB)	GPUs
BFloat16	109B * 2 bytes ~= 220GB	3 x 80GB
INT8/FP8	109B * 1 byte ~= 109GB **(-50%)**	2 x 80GB
INT4/FP4	109B * 0.5 byte ~= 55GB **(-75%)**	1 x 80GB

Inference Improvements

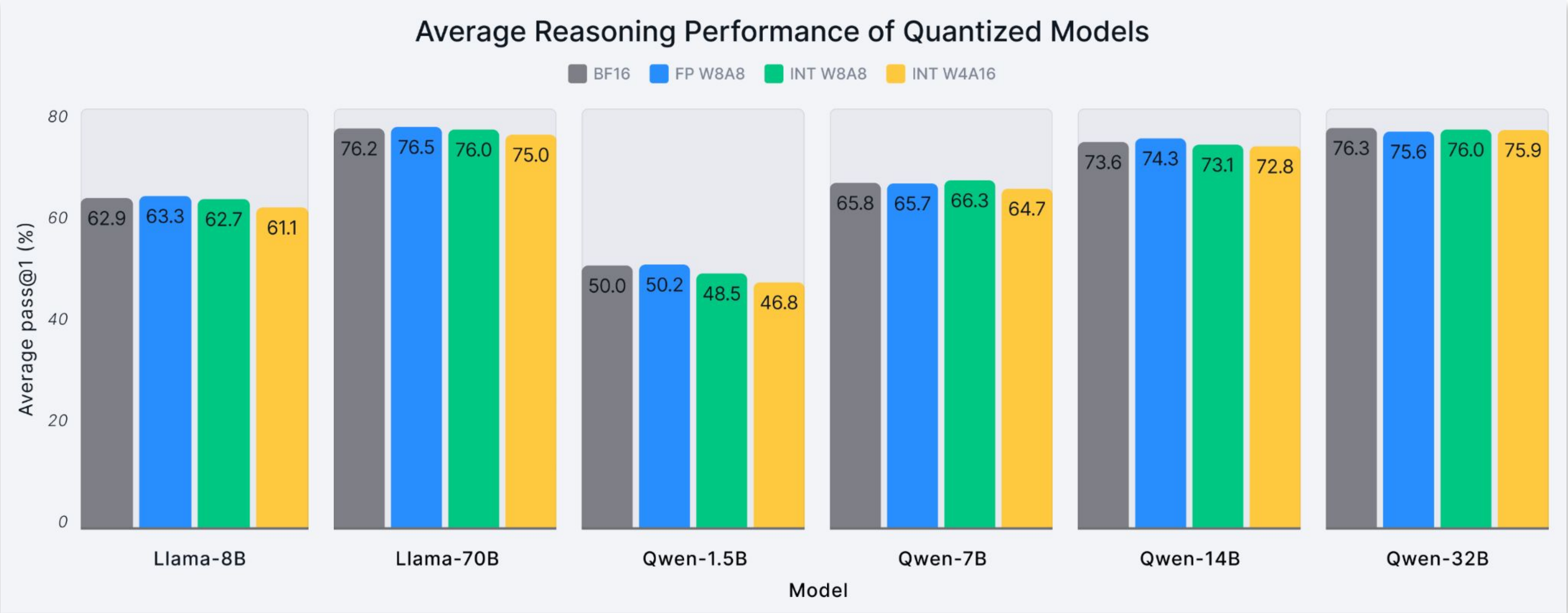
Performance! (Throughput and latency)





Impact on Model Accuracy

Minimal impact in most cases



[Kurtić, E., et al. (2025). "Quantized DeepSeek-R1 Models: Deployment-Ready Reasoning Models."]

How can I create my own optimized model?

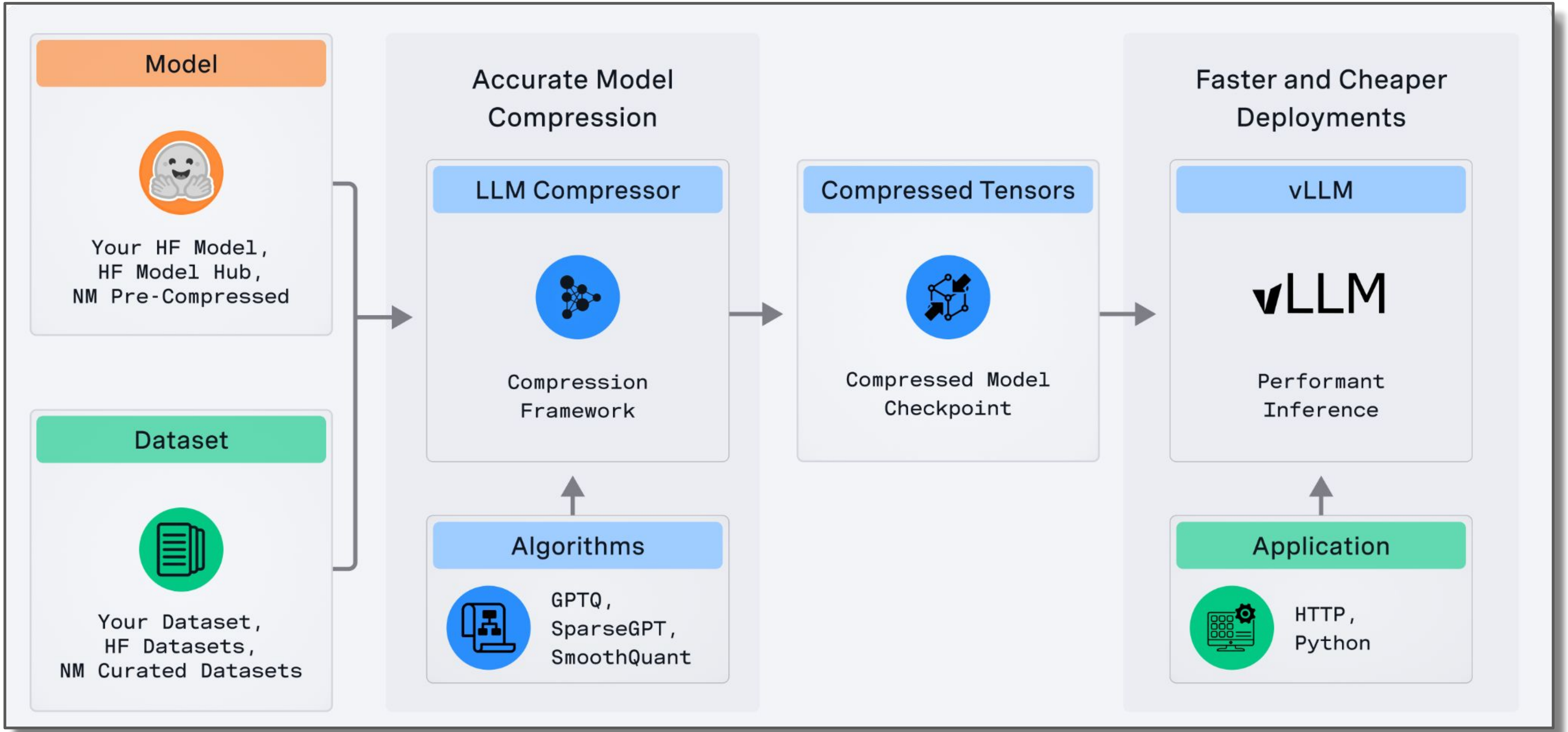
Red Hat

LLM Compressor

LLM Compressor is an **open-source** framework for applying SoTA research algorithms to **optimize models for inference in vLLM**

https://github.com/vllm-project/llm-compressor

LLM Compressor



https://github.com/vllm-project/llm-compressor

LLM Compressor

User Summary

- **For practitioners:** off-the-shelf validated models for use with vLLM
- **For developers:** easy to use recipes with optimized default configurations for all quantization formats supported in vLLM
- **For researchers:** fine-grained control
    - per-channel, per-tensor, per-group
    - symmetric/asymmetric
    - data-free and calibrated flows
    - activation reordering
    - dynamic/static activation quantization across precisions

How to use LLM Compressor with vLLM - Demo and Walkthrough

Red Hat

Where do I start?

Optimize your own models!

```
pip install llm-compressor
```

Use Red Hat optimized models!

huggingface.co/RedHatAI

Ask Questions!

Ask questions about LLM Compressor and model optimization

https://vllm-dev.slack.com/archives/C08MSER5NJ0

Get involved!

Check for issues with the `good first issue` label

https://github.com/vllm-project/llm-compressor/issues

The Red Hat AI team wants to hear from *you!*

Let us know your thoughts on Red Hat AI by
signing up to participate in research opportunities!

Scan the QR code or go to:
red.ht/uxfeedback

Have additional questions about [this product]?
Visit us at the Red Hat AI Booth

Red Hat

Red Hat Summit

Thank you

linkedin.com/company/red-hat

facebook.com/redhatinc

youtube.com/user/RedHatVideos

twitter.com/RedHat

Red Hat